INCH-POUND

MIL-I-82812(OS)
29 December 1989
SUPERSEDING
WS 22369A
21 July 1986
(See 6.8)

MILITARY SPECIFICATION

IGNITION STRANDS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements for two types of ignition strands composed of reactive material bound to glass fiber with an acrylic latex binder (see 6.1).

1.2 **Classification.** Ignition strands shall be of the following types (see 6.2):

a. Type I 0.035-inch diameter ignition strand
b. Type II 0.055-inch diameter ignition strand

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1376

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-A-192	Ammonium Perchlorate, Technical
MIL-P-217	Potassium Perchlorate
MIL-A-512	Aluminum Powder, Flaked, Grained, and Atomized
MIL-Y-1140	Yarn, Cord, Sleeving, Cloth, and Tape-glass
MIL-B-81705	Barrier Materials, Flexible, Electrostatic-protective, Heat Sealable

STANDARDS

FEDERAL

FED-STD-313	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of

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documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of the ignition strands shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials. The ignition strands shall be constructed of the materials specified herein.

3.2.1 Glass fiber. The glass fiber shall be in accordance with MIL-Y-1140, Class C, Form 1, Yarn No. ECD 225-1/2.

3.2.2 Binder. The binder used to bind the reactive material to the glass fiber shall be an acrylic latex binder (see 6.3.1).

3.2.3 Reactive material. The reactive material shall be formulated as specified in table I.

3.2.4 Diatomite. The diatomite utilized in the manufacture of the ignition strand shall be filtration grade, calcined.

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3.3 **Dimensions**. The ignition strands shall conform to the following dimensions:

a. Type I ignition strands: 0.035 ± 0.010 inch diameter

b. Type II ignition strands: 0.055 ± 0.010 inch diameter

TABLE I. Ignition strand reactive material formulation.

Material	Specification	Percentage, by weight
Ammonium Perchlorate 1/	MIL-A-192 (Grade A)	34.0 ± 5.0
Potassium Perchlorate 2/	MIL-P-217 (Grade A)	39.0 ± 5.0
Aluminum Flake	MIL-A-512 (Grade A) Type I, Class 1	21.0 ± 5.0
Diatomite	See 3.2.4	3.0 ± 2.0
Binder	See 3.2.2	3.0 +2.0 -0.0

1/ Weight percent includes 1.00 +1.00 -0.70 percent hydrophobic silica powder (see 6.3.2)

2/ Weight percentage includes 0.25 +0.75 -0.15 percent hydrophobic silica powder (see 6.3.2)

3.4 **Physical properties.**

3.4.1 **Caloric output**. The caloric output of the ignition strand shall be 1500 calories per gram (cal/g), minimum (see 4.5.2).

3.4.2 **Autoignition**. The ignition strand shall not ignite when exposed to a temperature of 177°C (350°F) for 2 hours (see 4.5.3).

3.5 **Safety.**

3.5.1 **Toxic products and formulations**. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.5.2 **Material Safety Data Sheets (MSDS)**. The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

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3.6 **Shelf life.** The ignition strand shall have a shelf life of 10 years after date of manufacture when stored in the original, desiccated, sealed containers at -20 to 130°F.

3.7 **Workmanship.** Workmanship shall be such that the ignition strand is uniform in appearance and free from visible contamination. Circumferential cracking of the ignition strand is inherent to the fabrication process and shall be acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the autoignition test specified in 4.5.3 in addition to the examinations and tests specified in table II. These tests shall be performed on ignition strand which is representative of the production material.

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4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

TABLE II. Quality conformance inspection.

Examination and Tests	Requirement	Method
Individual		
Visual examination	3.3 and 3.7	4.5.1
Inspection of packaging	Section 5	4.6
Sampling		
Caloric output	3.4.1	4.5.2

4.4.1 Lot. A lot shall consist of all the ignition strands manufactured in one continuous run, using the same lots of materials, made by the same processes and procedures, and subject to inspection at the same time.

4.4.1.1 **Sampling at manufacturing activity.** Sampling for quality conformance inspection at the manufacturing activity shall consist of a minimum of 2 feet of the ignition strand cut from both ends of the material. Noncompliance shall be cause for rejection of the lot.

4.4.1.2 **Sampling at contracting activity.** Sampling for quality conformance at the contracting activity shall consist of a minimum of 2 feet of strand cut from the outer end of the lot.

4.5 **Test methods.** The following test methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Visual examination shall be made to verify conformance to 3.3 and 3.7.

4.5.2 Caloric output. The caloric output of a 2.0 $\pm$ 0.4-gram sample shall be determined using a Parr Bomb Calorimeter, Model 1221 (or equivalent) in accordance with the manufacturer's manual number 120, except that no water shall be added to the bomb, and that the test shall be conducted at 25 atmospheres of helium or argon after three purges of the bomb with helium or argon at that pressure.

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4.5.3 **Autoignition**. A 1-inch (minimum) length of ignition strand shall be exposed to a temperature of 177 to 182°C (350 to 360°F) for 2 hours minimum. The ignition strand shall be placed in a container preconditioned to the required temperature and maintained at the temperature for the required time.

4.6 **Inspection of packaging**. The ignition strand containers and container markings shall be visually examined to verify compliance with Section 5.

5. **PACKAGING**

5.1 **Packaging**. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C.

5.1.1 **Preservation**. Preservation of the ignition strands shall be in accordance with MIL-P-116, submethod IA-8, with desiccant. The bag material shall conform to MIL-B-81705.

5.1.2 **Packing**. The ignition strand shall be packed to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.6. Containers in the same shipment shall be of the same size. The packing shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking**.

5.2.1 **Standard marking**. In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking**. In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Net weight of contents.

d. Shelf life expiration date.

e. Storage conditions.

f. Toxicity warning (if applicable).

g. Explosive classification.

NOTE: Material is class "C" explosive. Handle with care per 49 CFR.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The ignition strand specified herein is intended for use in the manufacture of ignition thin layer explosives for use in the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Type required (see 1.2).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

d. Whether first article inspection is required (see 3.1 and 6.4).

e. Material Safety Data Sheet required (see 3.5.2 and 6.5).

f. Packaging required, if different than 5.1.

g. Special marking, if other than as specified in 5.2.2.

h. Safety precautions required (see 6.6).

6.3 **Possible materials.** The following materials have found to meet the requirements of this specification. This list is given for information only.

6.3.1 **Binder.** Hycar 2671 manufactured by B.F. Goodrich, an acrylic latex binder, has been used in the ignition strand.

6.3.2 **Silica powder.** Aerosil R972 manufactured by Degussa Corporation, Peterboro, New Jersey, has been used in the ignition strand reactive material.

6.3.3 **Typical material.** Ignition strand manufactured by Explosive Technology, Fairfield, CA, has been found to meet this specification.

6.4 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding

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arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.8 **Supersedure information.** MIL-I-82812 incorporates the following engineering change proposal (ECP) and notice of revision (NOR):

ECP	NOR
MTA097 (8/22/89)	MTA097.1 (8/8/89)

Preparing Activity:
NAVY-OS
(Project 1376-N364)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-82812	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
IGNITION STRANDS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete. 198/290

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